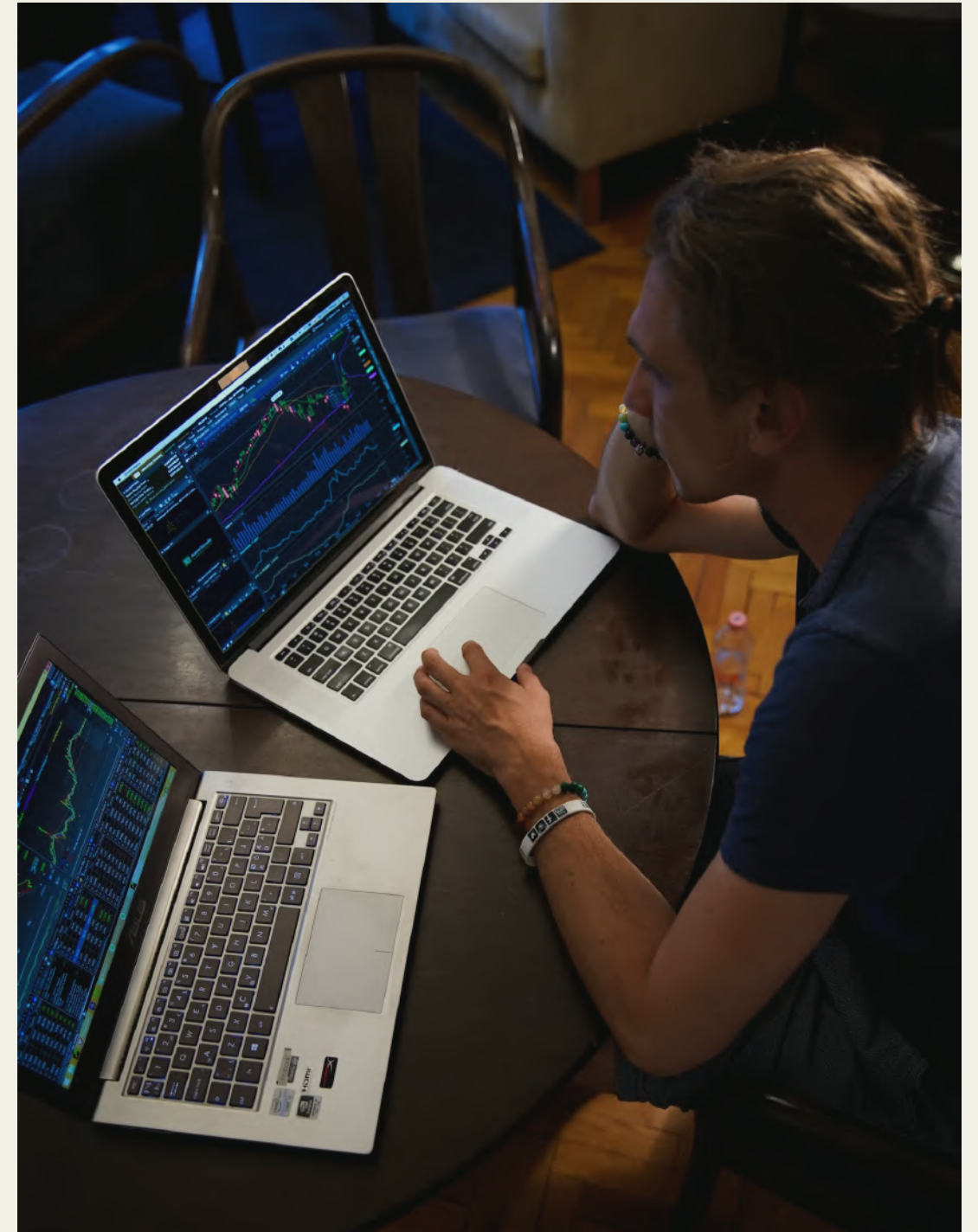


Customer Segmentation with Power BI

MARKET ANALYSIS

Empowering strategic decisions through customer segmentation insights



VISUAL INSIGHTS

Data Exploration

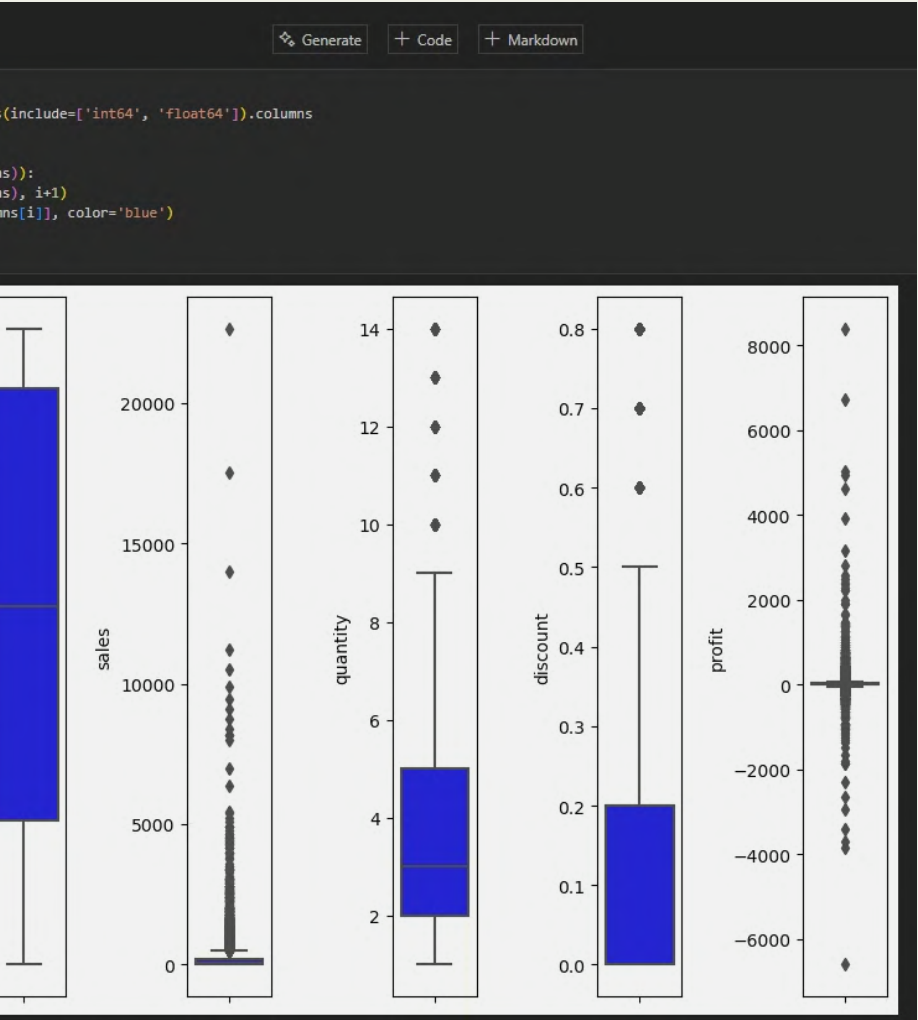
No missing value.

Have some outliers on sales, quantity, discount, and profit.
However, since this tables are business context, this could be because of a huge discount.

EDA

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column          Non-Null Count  Dtype
---  -
0   row_id           9994 non-null   int64
1   order_id         9994 non-null   object
2   order_date       9994 non-null   datetime64[ns]
3   ship_date        9994 non-null   datetime64[ns]
4   ship_mode        9994 non-null   object
5   customer_id      9994 non-null   object
6   customer_name    9994 non-null   object
7   segment          9994 non-null   object
8   country          9994 non-null   object
9   city             9994 non-null   object
10  state            9994 non-null   object
11  postal_code      9994 non-null   int64
12  region           9994 non-null   object
13  product_id       9994 non-null   object
14  category         9994 non-null   object
15  subcategory      9994 non-null   object
16  product_name     9994 non-null   object
17  sales            9994 non-null   float64
18  quantity         9994 non-null   int64
19  discount         9994 non-null   float64
20  profit           9994 non-null   float64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 1.6+ MB
```



RFM Segment Definition

RECENCY



Days since the last purchase (lower = better)

FREQUENCY



How often a customer purchased

MONETARY



How much they spent in total

Customer Segmentation

CHAMPIONS



High Recency, Frequency, and Monetary. (511 - 555)

LOYAL



High Frequency, moderate Recency and Monetary. (451 - 510)

POTENTIAL LOYALIST



Recent buyers with moderate frequency/spending. (351 - 450)

AT RISK



Previously good customers, but lately no purchased. (251 - 350)

LOST



No recent activity, low frequency and spending. (151 - 250)

UNCATEGORIZED



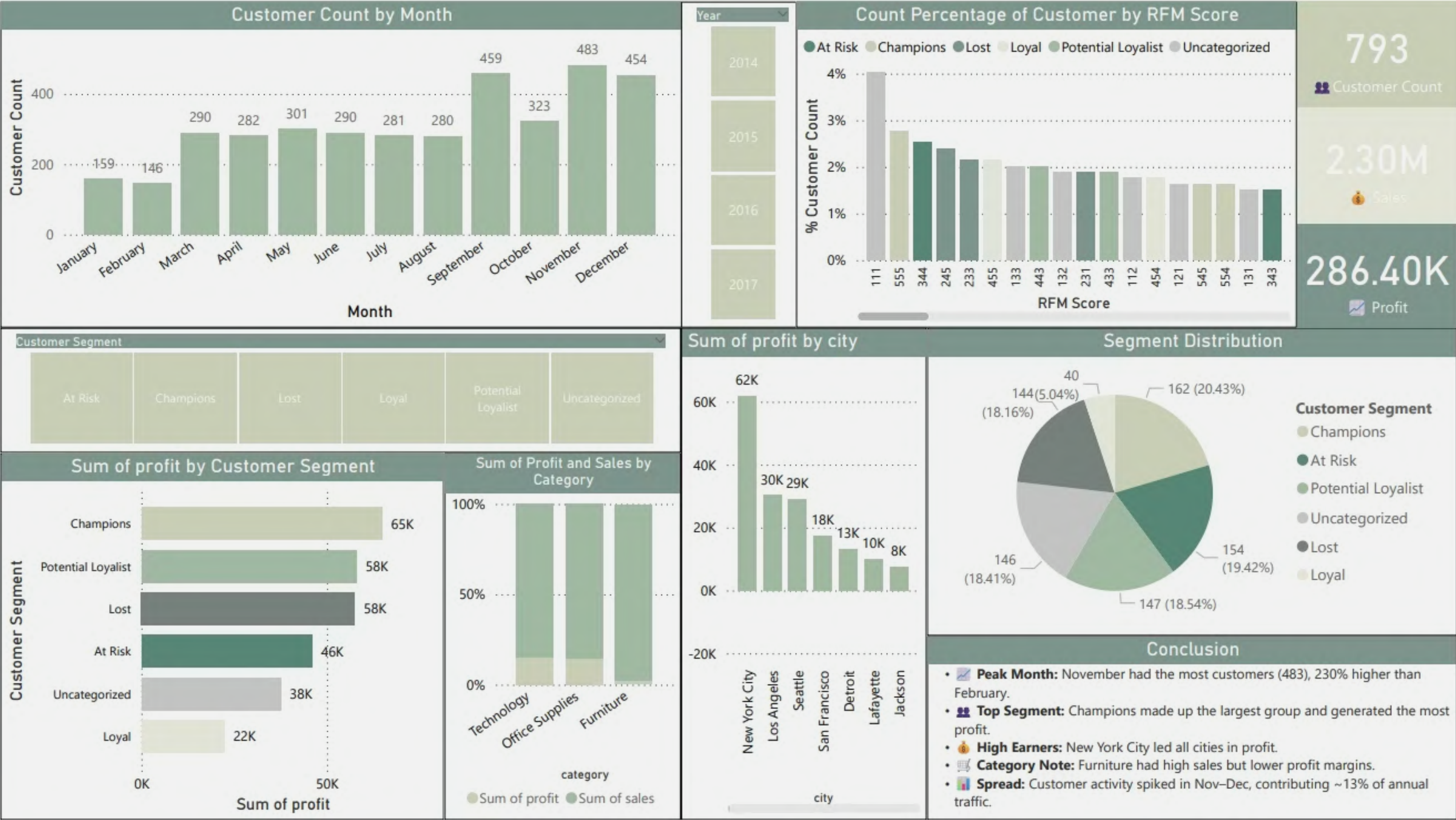
Doesn't fall clearly into any defined group. (≤ 150)

DATA VISUALIZATION

Importing insights from Power BI for clear analytics

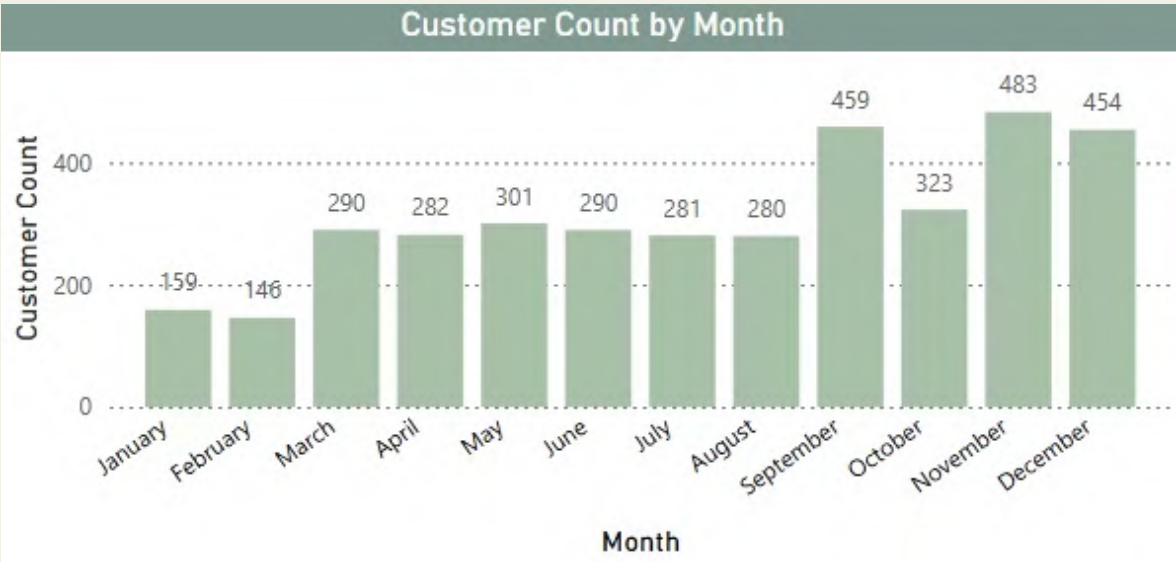


Power BI Data Visualization



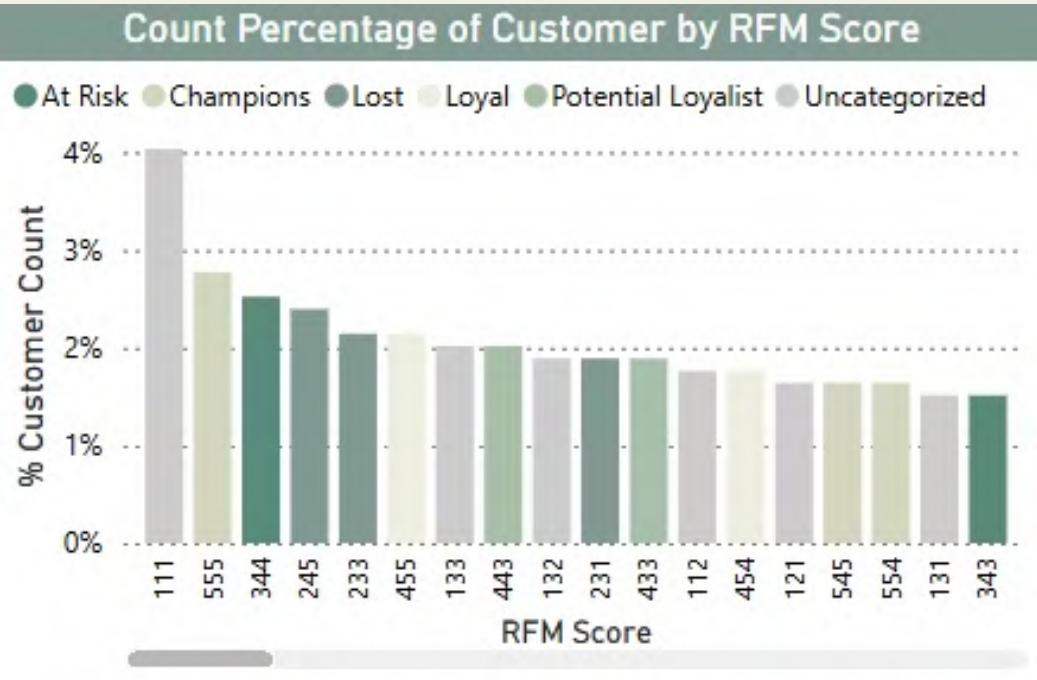
Insightful Charts

CUSTOMER COUNT BY MONTH



November had the highest customer count at **483**, which was **230.82% higher** than February (the lowest at 146). The top 3 months were **November, September, and December**

COUNT PERCENTAGE OF CUSTOMER BY RFM SCORE



Leveraging Power BI for data insights

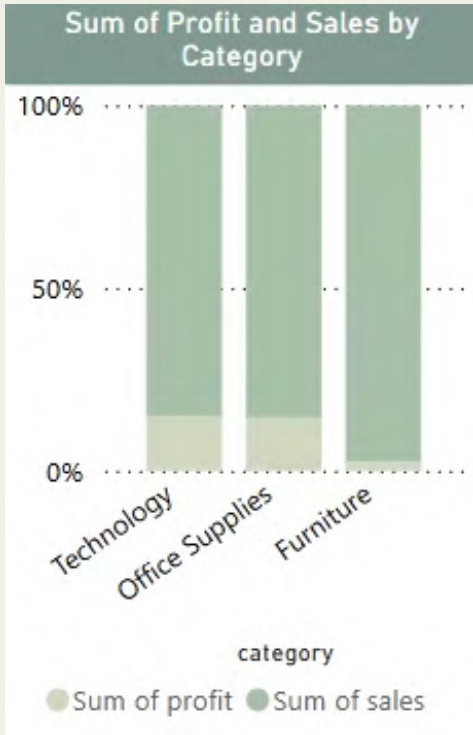
SUM OF PROFIT BY CUSTOMER SEGMENT



Champions contributed the most to profit (**65K**), followed by **Potential Loyalists** and **Lost**. Despite having fewer customers, **Loyal** and **Uncategorized** generated the least profit.

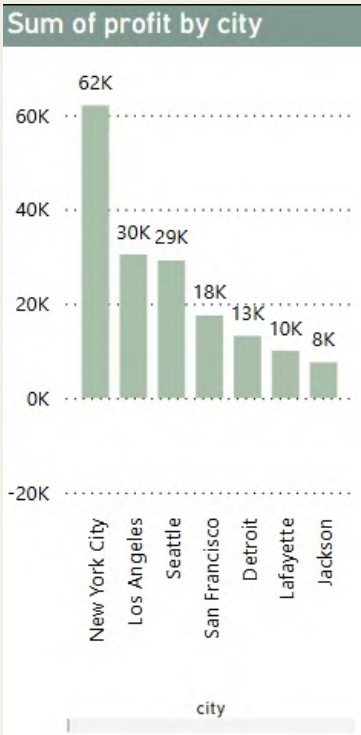
Insightful Charts

SUM OF PROFIT AND SALES BY CATEGORY



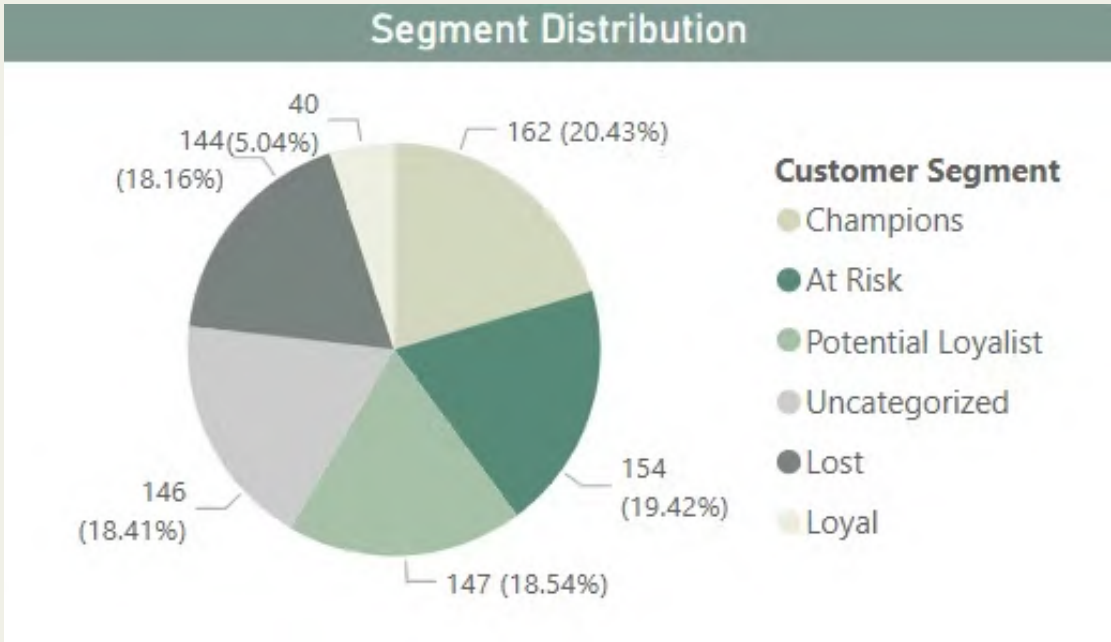
Although **Furniture** had high sales, it delivered lower profit margins compared to **Technology** and **Office Supplies**, possibly due to **higher average discounts**.

SUM OF PROFIT BY CITY



New York City generated the highest profit (**62K**), significantly outperforming others like **Los Angeles** and **Seattle**, indicating a key geographical target.

SEGMENT DISTRIBUTION



The largest group was **Champions** (20.43%), followed by **At Risk**, **Lost**, and **Potential Loyalists**. These segments help identify high-value customers and those needing attention.

Business Recommendations

PRIORITIZE CHAMPIONS



LAUNCH VIP PROGRAMS OR EARLY-ACCESS SALES TO RETAIN AND UPSELL THEM

NURTURE “POTENTIAL LOYALISTS”



OFFER VOLUME-BASED DISCOUNTS OR POINT-BASED REWARDS TO INCREASE VALUE.

WIN BACK "LOST" AND "AT RISK"



RUN REACTIVATION CAMPAIGNS VIA PERSONALIZED EMAILS OR ONE-TIME DEALS

INVESTIGATE PRICING AND DISCOUNT



FURNITURE HAS HIGH SALES VOLUME BUT LOWER RELATIVE PROFIT.
REVIEW PRICING AND DISCOUNT STRATEGIES TO IMPROVE MARGIN.

FOCUS ON HIGH PROFIT CITIES



ALLOCATE MORE MARKETING BUDGET OR INVENTORY TO THESE HIGH-PERFORMING REGIONS.

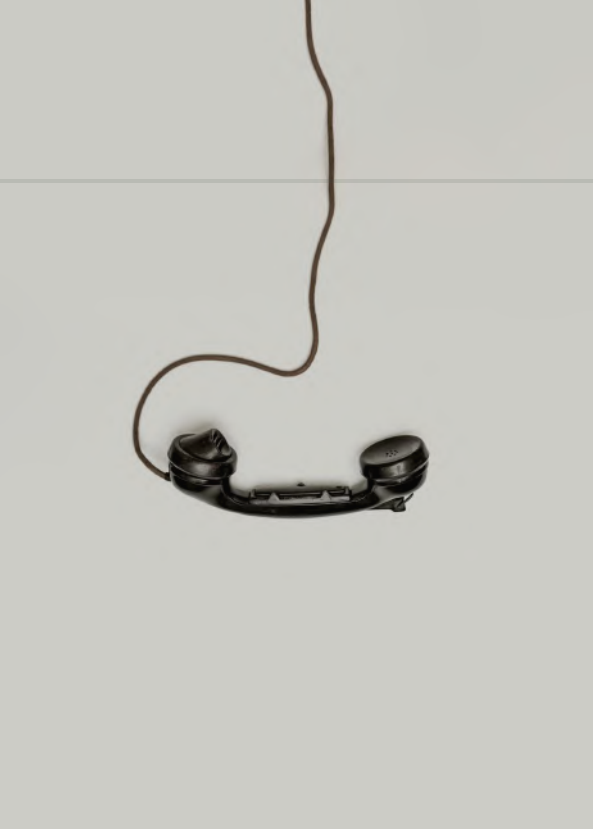
Appendices



EDA



Power BI



Contact Information

Phone

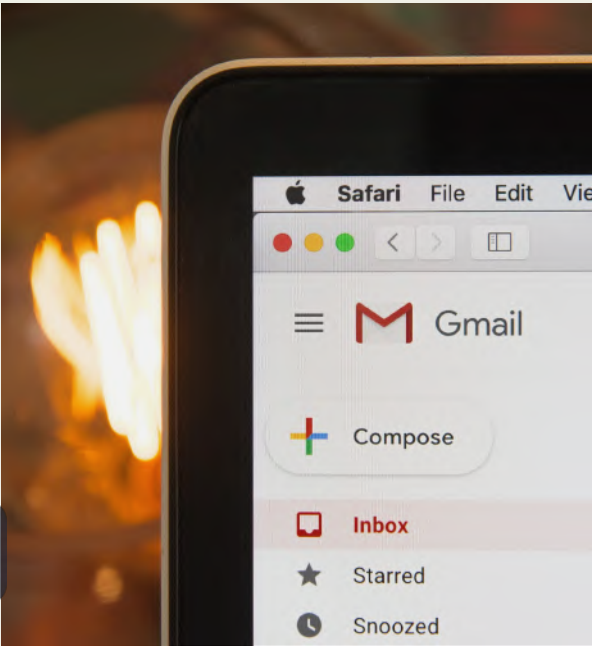
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